

```
R1
R1#sh ip dhcp pool

Pool HR :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 62
Leased addresses : 2
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
50.0.1.132 50.0.1.129 - 50.0.1.190 2
R1#sh ip dhcp binding
Bindings from all pools not associated with VRF:
IP address Client-ID/ Hardware address/ Lease expiration Type
User name
50.0.1.130 0100.5079.6668.6e Aug 05 2026 08:20 AM Automatic
50.0.1.131 0150.be84.086b.00 Aug 05 2026 08:21 AM Automatic
R1#sh ip int brief
Interface IP-Address OK? Method Status Prot
Gig0/0 50.0.1.129 YES manual up up
Gig0/1 50.0.1.193 YES manual up up
Gig0/2 unassigned YES TFTP down down
Gig0/3 unassigned YES TFTP down down
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 50.0.1.194 to network 0.0.0.0

S* 0.0.0.0/0 [1/0] via 50.0.1.194
C 50.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C 50.0.1.128/26 is directly connected, Ethernet0/0
L 50.0.1.129/32 is directly connected, Ethernet0/0
C 50.0.1.192/30 is directly connected, Ethernet0/1
L 50.0.1.193/32 is directly connected, Ethernet0/1
R1#
```

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R2#sh ip dhcp pool

Pool PRODUCTION :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 126
Leased addresses : 2
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
50.0.1.4 50.0.1.1 - 50.0.1.126 2

Pool SALES :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 2
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
50.0.0.7 50.0.0.1 - 50.0.0.254 2

R2#sh ip dhcp binding
Bindings from all pools not associated with VRF:
IP address Client-ID/ Hardware address/ Lease expiration Type
User name
90.0.0.5 0100.5079.6668.70 Aug 05 2026 07:55 AM Automatic
90.0.0.6 0150.9126.0870.00 Aug 05 2026 08:34 AM Automatic
90.0.1.2 0150.9cc3.086e.00 Aug 05 2026 06:37 AM Automatic
90.0.1.3 0100.5079.6668.6f Aug 05 2026 02:33 AM Automatic
R2#
```



Search



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R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

 50.0.0.0/8 is variably subnetted, 8 subnets, 5 masks
S    50.0.0.0/24 [1/0] via 50.0.1.198
C    50.0.1.0/25 is directly connected, Ethernet0/0
L    50.0.1.1/32 is directly connected, Ethernet0/0
S    50.0.1.128/26 [1/0] via 50.0.1.193
C    50.0.1.192/30 is directly connected, Ethernet0/1
L    50.0.1.194/32 is directly connected, Ethernet0/1
C    50.0.1.196/30 is directly connected, Ethernet0/2
L    50.0.1.197/32 is directly connected, Ethernet0/2
R2#sh ip int brief
Interface      IP-Address      OK? Method Status      Prot
ooc1
Ethernet0/0    50.0.1.1        YES manual up          up
Ethernet0/1    50.0.1.194      YES manual up          up
Ethernet0/2    50.0.1.197      YES manual up          up
Ethernet0/3    unassigned      YES TFTP  down        down
R2#
```



Search



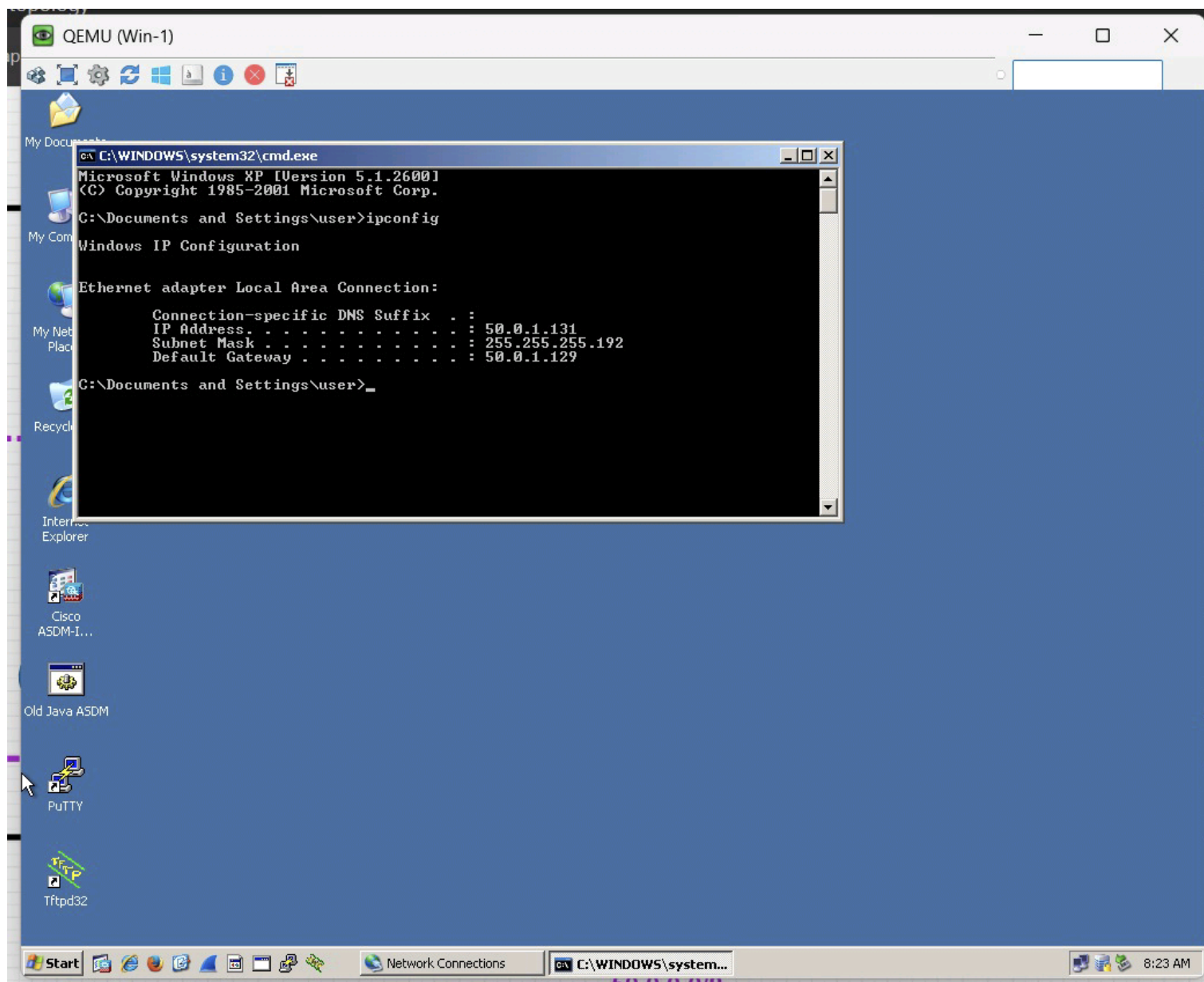
```
R3
R3(config-if)#
R3#sd
*Aug  4 08:41:53.970: %SYS-5-CONFIG_I: Configured from console by console
R3#sh ip int brief
Interface                IP-Address      OK? Method Status  Prot
Ethernet0/0              50.0.0.1        YES manual up      up
Ethernet0/1              50.0.1.198      YES manual up      up
Ethernet0/2              unassigned      YES TFTP  down    down
Ethernet0/3              unassigned      YES TFTP  down    down

R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 50.0.1.197 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 50.0.1.197
     50.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C     50.0.0.0/24 is directly connected, Ethernet0/0
L     50.0.0.1/32 is directly connected, Ethernet0/0
C     50.0.1.196/30 is directly connected, Ethernet0/1
L     50.0.1.198/32 is directly connected, Ethernet0/1

R3#sh ip helper-address
Interface      Helper-Address  VPN VRG Name  VRG State
Ethernet0/0    50.0.1.197     0   None        Unknown
R3#
```



VPC-1

VPC-1>

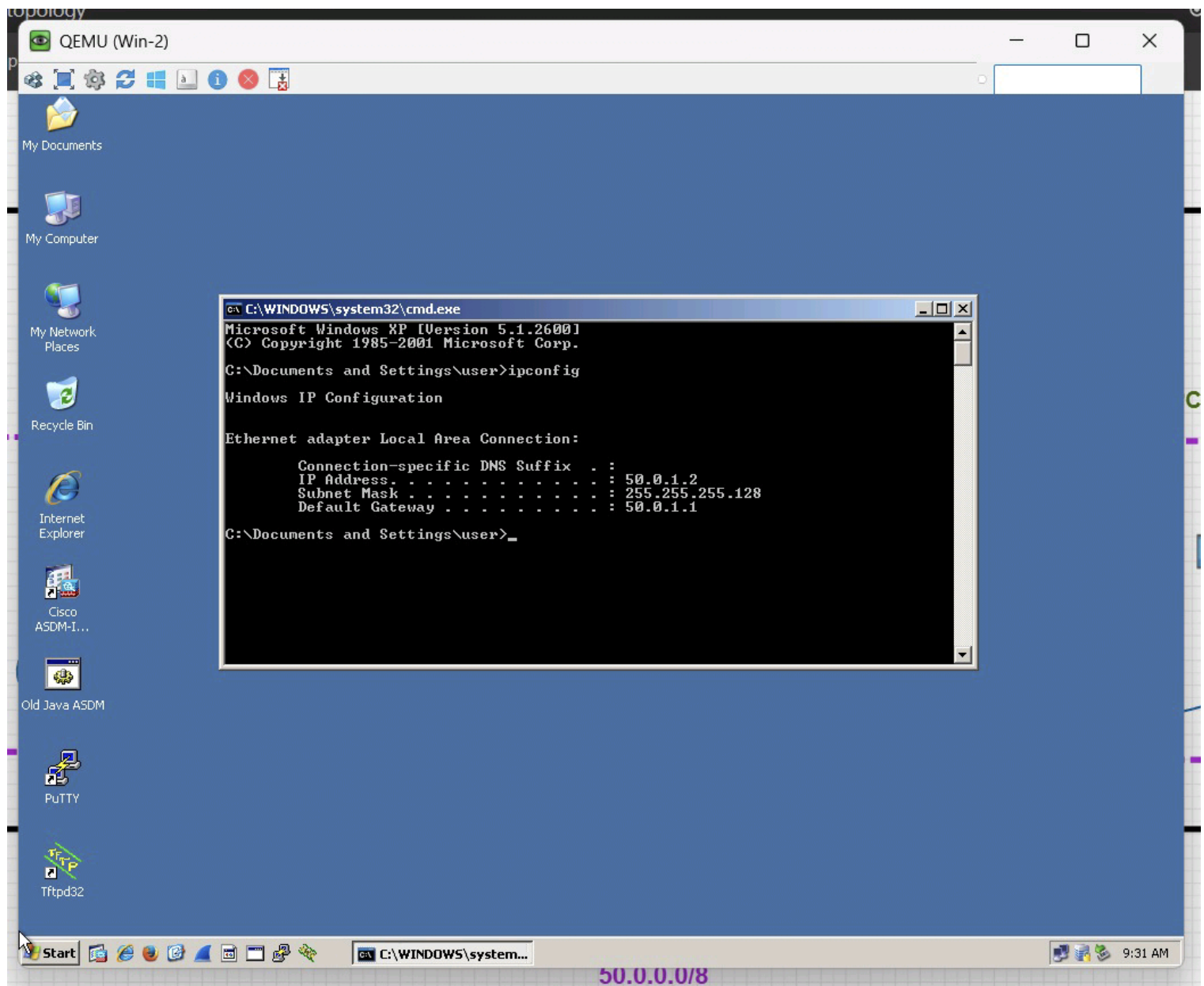
VPC-1>

VPC-1>

VPC-1> sh ip

NAME : VPC-1[1]  
IP/MASK : 50.0.1.130/26  
GATEWAY : 50.0.1.129  
DNS :  
DHCP SERVER : 50.0.1.129  
DHCP LEASE : 86186, 86400/43200/75600  
MAC : 00:50:79:66:68:6e  
LPORT : 20000  
RHOST:PORT : 127.0.0.1:30000  
MTU : 1500

VPC-1> █



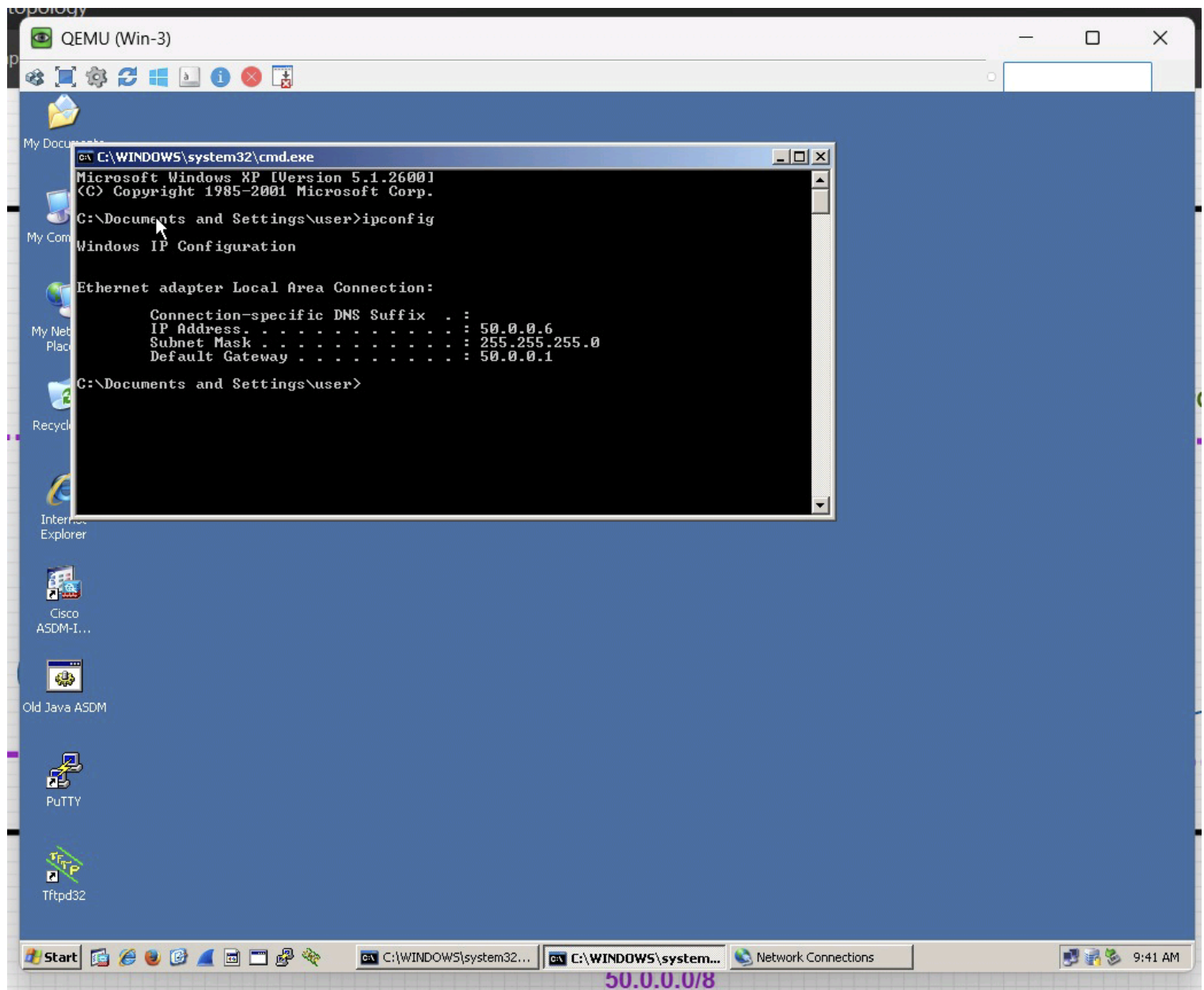
VPC-2

VPC-2> sh ip

NAME : VPC-2[1]  
IP/MASK : 50.0.1.3/25  
GATEWAY : 50.0.1.1  
DNS :  
DHCP SERVER : 50.0.1.1  
DHCP LEASE : 67425, 86400/43200/75600  
MAC : 00:50:79:66:68:6f  
LPORT : 20000  
RHOST:PORT : 127.0.0.1:30000  
MTU : 1500

VPC-2> █





VPC-3

ed  
Bad command: "pipe2 failed". Use ? for help.

VPC-3>

VPC-3> dhcp

DDORA IP 50.0.0.5/24 GW 50.0.0.1

VPC-3> sh ip

NAME : VPC-3[1]  
IP/MASK : 50.0.0.5/24  
GATEWAY : 50.0.0.1  
DNS :  
DHCP SERVER : 50.0.1.197  
DHCP LEASE : 83616, 86400/43200/75600  
MAC : 00:50:79:66:68:70  
LPORT : 20000  
RHOST:PORT : 127.0.0.1:30000  
MTU : 1500

VPC-3> █

Win-1

VPC-1

Win